Generative AI and RAG – Test Document

1. Artificial Intelligence

Artificial Intelligence (AI) is a field of computing focused on building systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing patterns, and making decisions.

2. Machine Learning

Machine Learning (ML) is a branch of AI in which systems learn patterns from data. Supervised learning uses labeled examples, while unsupervised learning finds patterns in data without predefined labels.

3. Deep Learning

Deep Learning uses neural networks with multiple layers to learn complex patterns. It is commonly used for image recognition, speech processing, and natural language tasks.

4. Generative AI

Generative AI refers to models that can create new content such as text, images, audio, and code. Large language models are a common example of Generative AI systems for producing and understanding text.

5. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. A user question is converted into an embedding, relevant document chunks are retrieved from a vector database, and the retrieved context is provided to a language model to generate an answer.

6. Embeddings

An embedding represents text as a numerical vector. Text with similar meaning can have similar vector representations, allowing a system to perform semantic search rather than depending only on exact keyword matches.

7. Vector Databases

A vector database stores vector representations and allows similarity searches. In a RAG application, a vector database can be used to retrieve the document chunks that are most relevant to a user's question.

8. Test Facts

The sample RAG application uses Sentence Transformers to create embeddings, ChromaDB as the vector database, and Groq as the language-model service. The application can retrieve information from multiple uploaded PDF documents.